Type MS · Explain validity problems and fixes in investigations on people

Answer 1 of 6 · Question 8 part b.ii · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

8 (a)(i)	• high (blood) pressure (in the glomerulus); • small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule);	Do not allow protein	1 1
(ii)	Any four from: • lower concentration of glucose/oxygen in blood in renal vein; • used in respiration; • less/no urea in renal vein; • urea excreted in urine; • more carbon dioxide in renal vein; • carbon dioxide produced in respiration;	Allow reverse argument throughout	Max 4
(b) (i)	quantity of salt (in food);		1
(ii)	Any two from: • different foods used (which may affect the amount of urea produced/amount of water reabsorbed); • different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance); • the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced);		Max 2
(iii)	use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content);	Allow valid alternatives	1

Total question 8 = 10 marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 1 of 6 · Question 8 part b.ii · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 2 of 6 · Question 8 part b.iii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

8 (a)(i)	• high (blood) pressure (in the glomerulus); • small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule);	Do not allow protein	1 1
(ii)	Any four from: • lower concentration of glucose/oxygen in blood in renal vein; • used in respiration; • less/no urea in renal vein; • urea excreted in urine; • more carbon dioxide in renal vein; • carbon dioxide produced in respiration;	Allow reverse argument throughout	Max 4
(b) (i)	quantity of salt (in food);		1
(ii)	Any two from: • different foods used (which may affect the amount of urea produced/amount of water reabsorbed); • different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance); • the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced);		Max 2
(iii)	use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content);	Allow valid alternatives	1

Total question 8 = 10 marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 2 of 6 · Question 8 part b.iii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 3 of 6 · Question 5 part d · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Question number			
5 (a)	• addition of figures = 1690; • divide by 10 = 169;		2
(b)	makes it faster/decreases reaction time;		1
(c)	• caffeine has to be absorbed; • into blood; • transported to nervous system/brain;	Ignore body	3
(d)	• can't control what is in food/drink; • could contain caffeine/other factors that would affect result/to ensure caffeine is the cause;	Ignore coffee	2
(e)	185 ± 5 mm;		1 Total 9

Type MS · Explain validity problems and fixes in investigations on people

Answer 3 of 6 · Question 5 part d · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 4 of 6 · Question 5 part c.i · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Explain validity problems and fixes in investigations on people

Answer 4 of 6 · Question 5 part c.i · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 5 of 6 · Question 6 part b.iv · 2 marks

2024 JUN · Paper 02 (regional R) · find part b.iv in this mark scheme extract

6	(a)	• reduced blood flow to the heart/cardiac muscle/coronary arteries; • less oxygen to heart/cardiac muscle/tissue; • less (aerobic) respiration/less energy for contraction; • (cardiac) muscle/cells die; • (leading to) heart attack/heart disease;		(4)
	(b) (i)	As the use of statins increases the level of blood cholesterol decreases;	allow inversely proportional	(1)
	(ii)	1988 = 31; 2005 = 17; 1 : 1.8/1 : 0.54/1:0.548/1:0.55;	ecf for incorrect graph readings. if use 18:17 ratio is 1:1.0588 (accept 1:1.059, 1.06, 1.1) full marks for correct final answer	(3)
	(iii)	• select sample of people; • record the number of people taking statins/not taking statins; • take blood samples; • measure the concentration/level of cholesterol (in blood); • repeat every year/between 1988-2008;		(5)
	(iv)	any two from the following: • gender/age not taken into account; • sample size maybe too small; • reference to a named lifestyle not taken into account e.g. diet, level of activity/smoking; • genetic predisposition/diseases not taken into account;		(2)

Total for Question 6 = 15 marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 6 of 6 · Question 5 part a.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

number				
5	(a) (i)	any two from • to allow the heart rate to return to normal; • allow respiration/breathing rate/body temperature to return to normal; • to allow sweat to evaporate;		2
	(ii)	any three from • same time for exercise; • same type/intensity of exercise; • wear same clothing for each test; • measure sweat from same area;		3
	(b) (i)	• axes labelled with units with independent variable on x-axis; • suitable scale; • both sets of data plotted correctly;; • suitable lines; • lines labelled;		6
	(ii)	any five from • exercise increases muscle activity; • more energy/ATP required; • increased respiration; • heat produced as by product; • lost from body; • by increased sweating; • increase in room temperature increases rate of sweating;		5

Total 16 marks

Type MS · Explain validity problems and fixes in investigations on people

Answer 6 of 6 · Question 5 part a.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks